

Joan Vila Orús

Applied AI & Machine Learning Engineer

LLM Systems | Computer Vision | 3D/VR | Python & C++

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PROFILE

Computer Engineering graduate from UPC-FIB with experience in clinical NLP, LLM systems, RAG, FastAPI backends, Unity/VR and 3D graphics. Developed privacy-sensitive NLP workflows at Hospital Clínic de Barcelona and built a VR assistant grounded in 3D scene context. Interested in computer vision, 3D perception and embodied AI.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern

Hospital Clínic de Barcelona

Sep 2025 – Mar 2026

Barcelona, Spain

- Developed Python NLP pipelines for clinical-document anonymization and sensitive-information detection in a privacy-sensitive healthcare environment.
- Evaluated Transformer-based models with PyTorch and benchmarking workflows, focusing on anonymization quality, reliability and reproducibility.
- Built data-processing and evaluation scripts, and explored LLM-based synthetic-data generation for internal experimentation.

Activity Coordinator & Resident Assistant

RESA Student Residences

Sep 2024 – Jul 2025

Barcelona, Spain

- Coordinated activities for a residence of 165+ students, managing planning, communication, logistics and execution.
- Managed annual activity budgets and collaborated with coordinators across 40+ student residences nationwide.

Basketball Coach

CB Peña Fragatina

2021 – 2023

- Planned, coordinated and led basketball training sessions, adapting activities to the players' needs and skill levels.
- Provided clear instructions, encouraged active participation and promoted teamwork among players.

SELECTED PROJECTS

AI-Assisted VR Simulation with Spatial Context

Unity, C#, Python, FastAPI, RAG, Llama 3, Whisper, Piper

- Built a voice-based AI guide for a VR anatomy simulation that adapts its response to the 3D object the user is pointing at.
- Designed a FastAPI backend connecting Unity with local/online LLM inference, speech-to-text, text-to-speech and retrieval-augmented generation.
- Added Cross-Encoder reranking and evaluated the prototype with 13 users on Meta Quest 2, analysing latency, feedback and reliability.

BlindNav — Mobile Visual Assistance Prototype

Python, OpenCV, YOLOv8

- Built an early proof of concept for visual assistance using real-time object detection and audio feedback.
- Designed as a first step toward smart-glasses and assistive computer-vision applications.

AI Performance Coach — Adaptive Training Assistant

Work in Progress | Python, FastAPI, LangChain, LLM APIs, RAG

- Developing an AI training assistant that generates adaptive workout plans from user goals, constraints and progression rules using LLM reasoning.

EDUCATION

B.Sc. Computer Engineering — Computing Specialization

Universitat Politècnica de Catalunya (UPC-FIB)

2022 – 2026

Barcelona, Spain

Bachelor Thesis: *Enhancing a Virtual Reality Simulation with an AI-Assisted Interactive Guide*

Erasmus+ Exchange — Computer Engineering

Sapienza Università di Roma

Feb 2026 – Jul 2026

Rome, Italy

LANGUAGES

Spanish: Native | Catalan: Native | English: B2 / Professional working proficiency | Italian: Medium